Physics Study Buddy

AI-Powered Agentic Study Assistant for B.Tech Physics

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1. Problem Statement

B.Tech students frequently struggle with physics concepts, formulas, and numerical problems outside classroom hours when no tutor is available. Existing resources require browsing multiple pages with no conversational interface. A 24/7 intelligent assistant is needed that explains curriculum topics faithfully, computes numerical answers on demand, and remembers the student across a session — without hallucinating formulas or constants.

Success Criteria

- 90%+ of curriculum questions answered faithfully (faithfulness ≥ 0.7).
- Correctly refuses out-of-scope questions and corrects false premises.
- Physics calculator handles numerical queries with explicit values accurately.
- Student name remembered across a multi-turn conversation via MemorySaver.

2. Solution & Features

The Physics Study Buddy is a LangGraph StateGraph with 8 nodes. It retrieves accurate content from a 12-document ChromaDB knowledge base, generates grounded answers using Groq LLaMA-3.3-70b, self-evaluates faithfulness before delivery, and uses a safe Python calculator for numerical queries.

Graph Architecture — 8 Nodes

Node	Responsibility
`memory_node`	Appends question to history; sliding window (last 6 msgs); extracts student name via regex.
`router_node`	LLM classifies question → retrieve / tool / memory_only (single-word output).
`retrieval_node`	Embeds query; queries ChromaDB for top-3 chunks by cosine similarity.
`skip_node`	Returns empty context for memory_only — bypasses vector store.
`tool_node`	LLM extracts math expression; safe eval() — math module only, no builtins. Never raises.
`answer_node`	Grounded system prompt; generates answer from context + tool result.
`eval_node`	Scores faithfulness 0–1 via fast LLM; routes back if < 0.7 (MAX_EVAL_RETRIES = 2).
`save_node`	Appends answer to messages history → END.

Six Mandatory Capabilities

- **LangGraph StateGraph (8 nodes)** — conditional edges after router & eval; CapstoneState TypedDict (10 fields) defined first.
- **ChromaDB RAG (12 docs)** — all-MiniLM-L6-v2 embeddings; top-3 cosine retrieval; tested before graph assembly.
- **MemorySaver + thread_id** — persistent multi-turn memory; sliding window (msgs[-6:]) prevents token overflow.
- **Self-reflection eval node** — faithfulness scoring with retry loop; refusal phrases auto-score 1.0.
- **Physics Calculator tool** — LLM-extracted expression; restricted eval() with math module only; no exceptions raised.
- **Streamlit deployment** — @st.cache_resource; st.session_state; sidebar with topic list; New Conversation reset.

3. Screenshots

The Streamlit UI screenshot below shows a live 3-turn session. In Turn 1 the student introduces themselves as Aryan. The agent greets them by name. In Turn 3 (summary request) the agent recalls the name unprompted using the memory_only route — confirming persistent cross-turn memory. Faithfulness: 1.00 and route are displayed in the caption below each response.

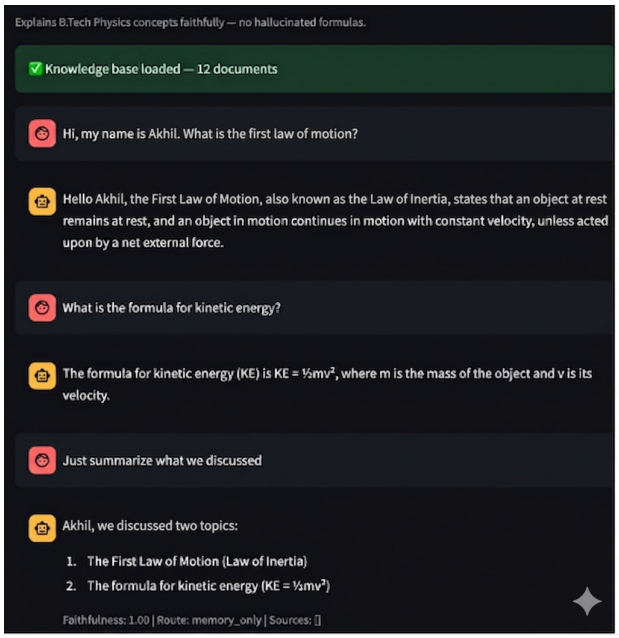

Figure 1: Physics Study Buddy — 3-turn Streamlit session. Agent greets 'Hello Akhil' (Turn 1, route: retrieve) and recalls name unprompted (Turn 3, route: memory_only, Faithfulness: 1.00).

4. Tech Stack

Technology	Version / Model	Role
LangGraph	Latest	StateGraph, conditional edges, MemorySaver checkpointing
ChromaDB	In-memory	Vector store for 12 Physics documents

SentenceTransformer	all-MiniLM-L6-v2	Dense embeddings — documents and queries
Groq LLaMA	3.3-70b-versatile	Primary LLM — routing, answer generation
Groq LLaMA	3.1-8b-instant	Eval LLM — faithfulness scoring (rate-limit efficient)
Streamlit	Latest	Chat UI — cache_resource, session_state, sidebar
RAGAS	0.2+	Automated eval: faithfulness, relevancy, precision
Python math stdlib	Built-in	Safe expression evaluation in calculator tool

5. Unique Points

- **Router-Level Safety:** Router prompt explicitly detects physically impossible values (e.g., speed of light = 1000 m/s) and routes to 'retrieve' instead of 'tool' — preventing silent computation on false data. This turned red-team test RT2 from FAIL to PASS (faithfulness 1.00).
- **Self-Reflection Eval Gate:** eval_node uses a secondary fast LLM to score faithfulness 0–1 and loop back to answer_node with escalated grounding if score < 0.7. Refusal phrases auto-score 1.0. MAX_EVAL_RETRIES = 2 prevents infinite loops.
- **Safe Physics Calculator:** eval() runs with restricted namespace (__builtins__ = {}, math module only). No arbitrary code execution. All errors returned as strings — graph never crashes on malformed expressions.
- **Zero OpenAI Dependency:** Full pipeline uses Groq free-tier LLaMA + local HuggingFace SentenceTransformer. Fully reproducible with only a free Groq API key.
- **RAGAS Evaluation Without OpenAI:** Configured with LangchainLLMWrapper (Groq) + HuggingFaceEmbeddings — scoring: Faithfulness 0.920 | Answer Relevancy 0.981 | Context Precision 1.000.

Metric	Score	Status
Faithfulness	**0.920**	✓ PASS
Answer Relevancy	**0.981**	✓ PASS
Context Precision	**1.000**	✓ PASS

Test Results — 11 / 11 Passed | Avg Faithfulness: 0.94

#	Test	Route	Faith.	Result
1	Newton's three laws	retrieve	0.90	✓ PASS
2	Projectile / 45° range	retrieve	0.90	✓ PASS
3	Work-energy theorem	retrieve	0.90	✓ PASS
4	Pendulum period	retrieve	0.90	✓ PASS
5	First Law of Thermodynamics	retrieve	0.90	✓ PASS
6	Snell's Law & TIR	retrieve	0.90	✓ PASS
7	De Broglie wavelength	retrieve	0.90	✓ PASS
8	KE of 3 kg at 4 m/s (calc)	tool	1.00	✓ PASS

9	KE value recalled (memory)	memory_only	1.00	✓ PASS
RT1	Biryani recipe (OOS)	retrieve	1.00	✓ PASS
RT2	Light at 1000 m/s (false)	retrieve	1.00	✓ PASS

6. Future Improvements

- **Textbook PDFs as Knowledge Base:** Replace hand-written summaries with HC Verma / NCERT chapters via PyPDF2 (200-word overlapping chunks). Current summaries miss derivation-heavy queries like 'derive $v^2 = u^2 + 2as$ step by step'.
- **Hybrid BM25 + Dense Retrieval:** Add rank_bm25 alongside ChromaDB for exact formula token matching (v^2, $\sin\theta$, ΔU) while dense vectors handle semantic queries.
- **LaTeX Formula Rendering:** Use st.latex() in Streamlit to render mathematical notation properly instead of plain-text approximations.
- **Expanded Topic Coverage:** Add electromagnetism, quantum mechanics, nuclear physics. Difficulty filtering: beginner / intermediate / advanced response modes.
- **Persistent ChromaDB Storage:** Replace in-memory store with local disk persistence — eliminates re-embedding on each startup and supports textbook-scale content.
- **Interactive Quiz Mode:** Agent generates MCQs from the knowledge base, tracks score, and adapts difficulty — transforming the tool from Q&A to active learning.

7. Knowledge Base (12 Documents)

12 hand-authored B.Tech-level documents (200–400 words each), one topic per document, with full formulas, worked examples, and derivations. Embedded with all-MiniLM-L6-v2; top-3 retrieval.

`doc_001` Newton's Laws of Motion	`doc_002` Kinematics — Equations of Motion
`doc_003` Work, Energy & Power	`doc_004` Simple Harmonic Motion
`doc_005` Thermodynamics	`doc_006` Electrostatics — Coulomb's Law
`doc_007` Current Electricity	`doc_008` Optics — Reflection, Refraction, Lenses
`doc_009` Modern Physics — Photoelectric & Bohr	`doc_010` Gravitation — Newton & Orbits
`doc_011` Rotational Motion — Torque & Inertia	`doc_012` Waves & Sound